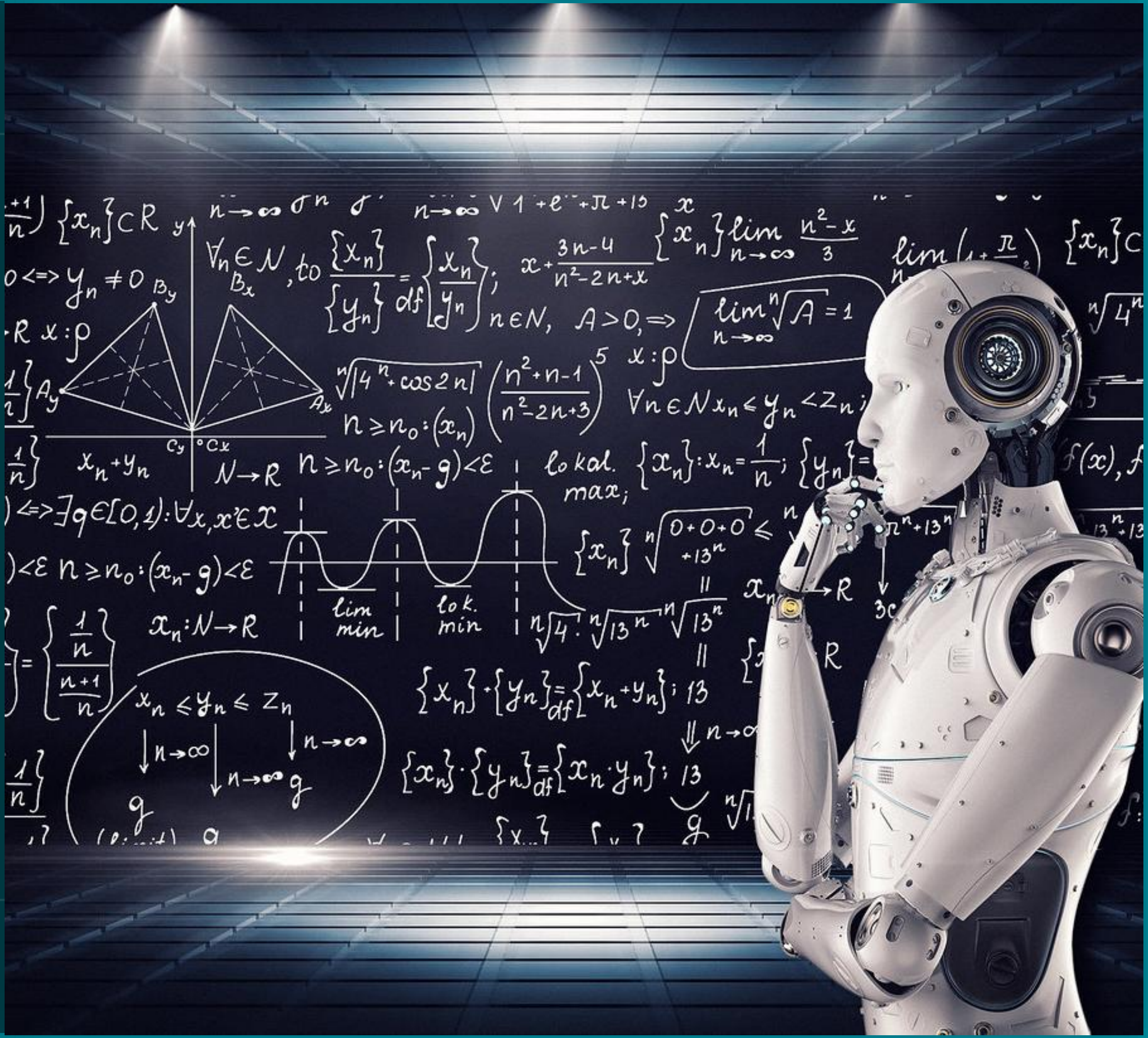
Introduction to Machine Learning

Introduction to Supervised Learning



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Machine Learning Types

Supervised Learning

Unsupervised Learning

Reinforcing Learning

Supervised Learning

Input   Output

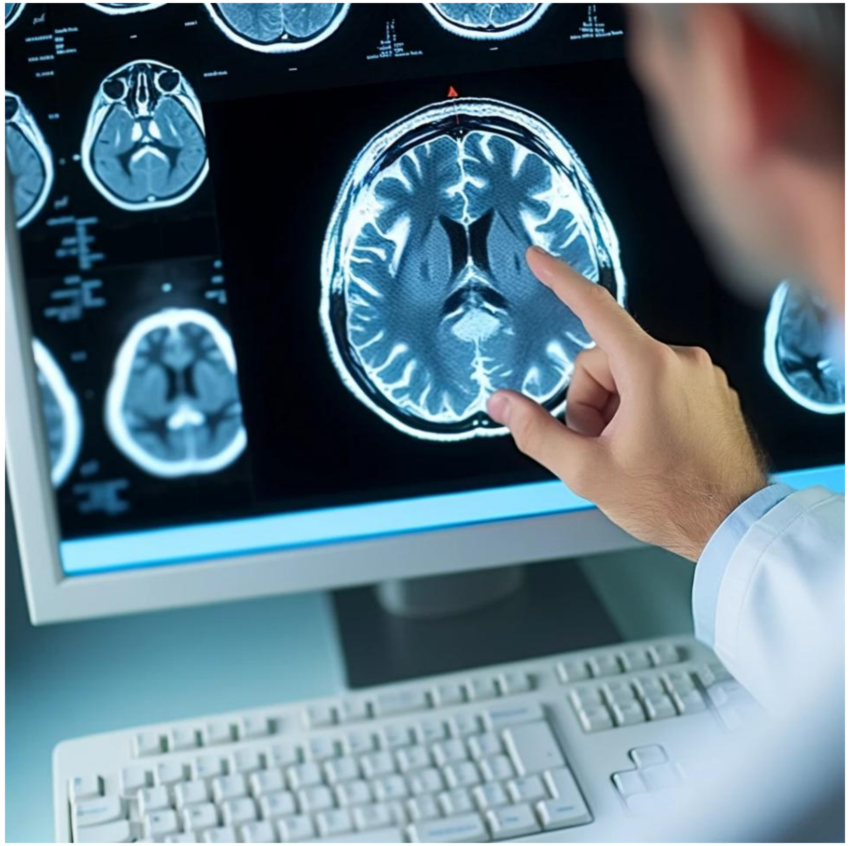
$$X \xrightarrow{\quad f(X) \quad} Y$$

Clinical applications of supervised learning

Input *X*: CT scans

Output *Y:* Tumor presence

Application *f(X)*: Image analysis



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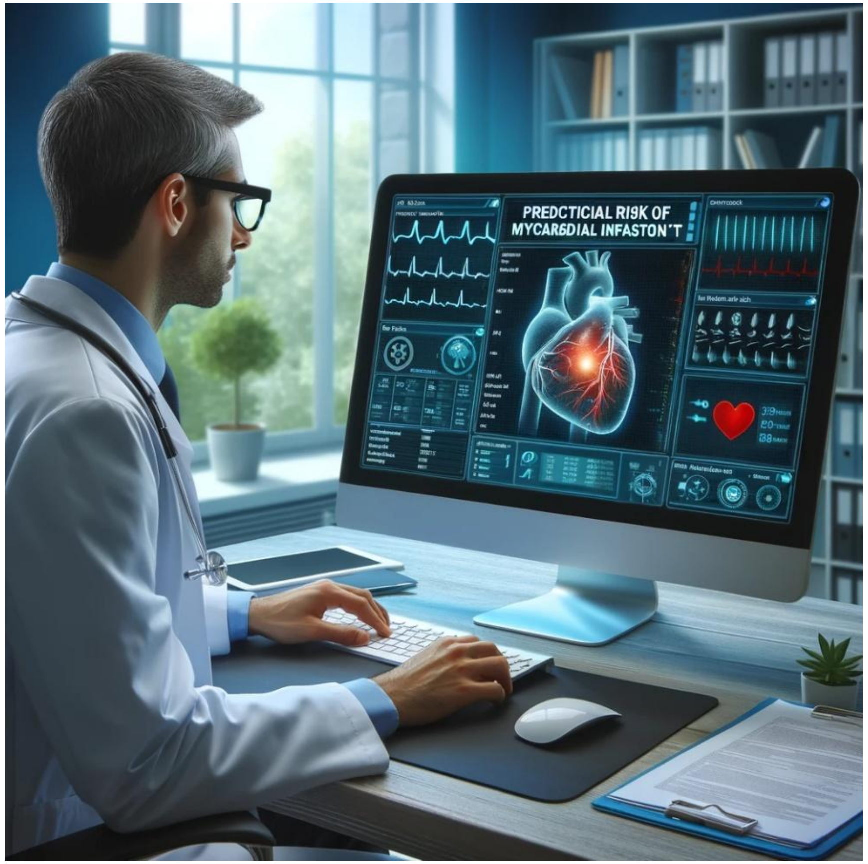
Clinical applications of supervised learning

Input X: Clinical data

Output Y: Myocardial infarction

Application f(X): Clinical risk prediction

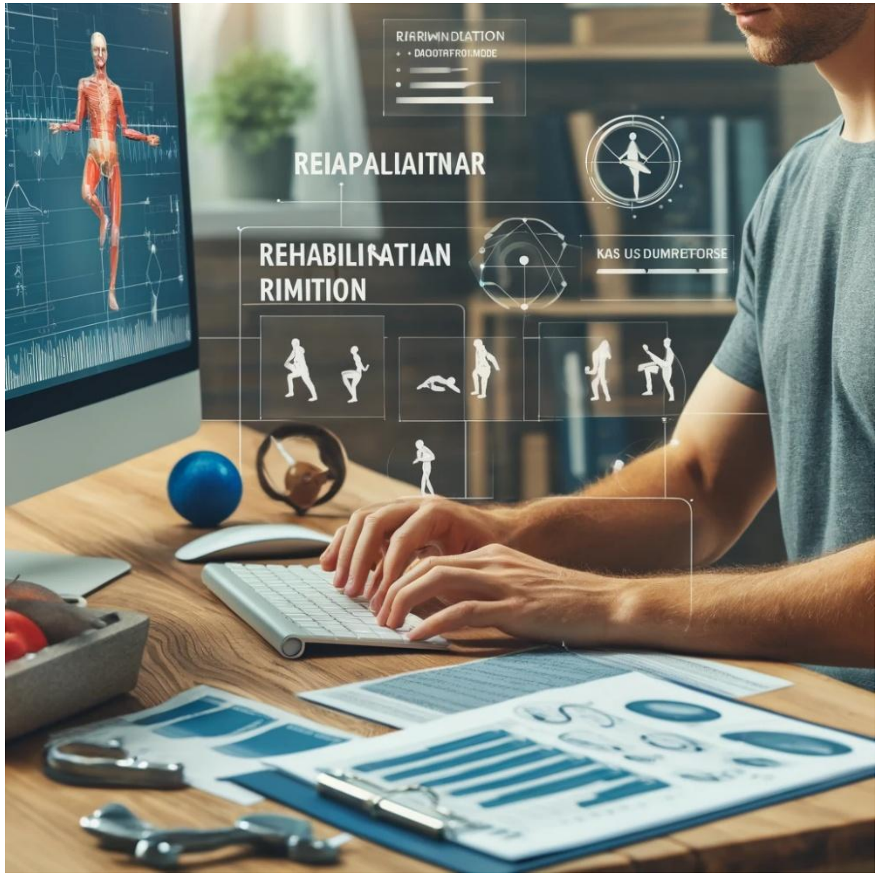


Clinical applications of supervised learning

Input X: Physical therapy assessment

Output Y: Improvement in physical function

Application f(X): Planning rehabilitation regimens



Clinical applications of supervised learning

Input X: Chemotherapy, patient data

Output Y: Tumor growth

Application f(X): Predict tumor response

Why estimate *f*(*X*)?

- Prediction

$$\hat{Y} = \hat{f}(X) + \varepsilon$$

- Inference

$$Y = f(X) + \varepsilon$$

How to estimate *f*(*X*)?

(x_1,y_1)

(x_3,y_3)

(x_2,y_2)

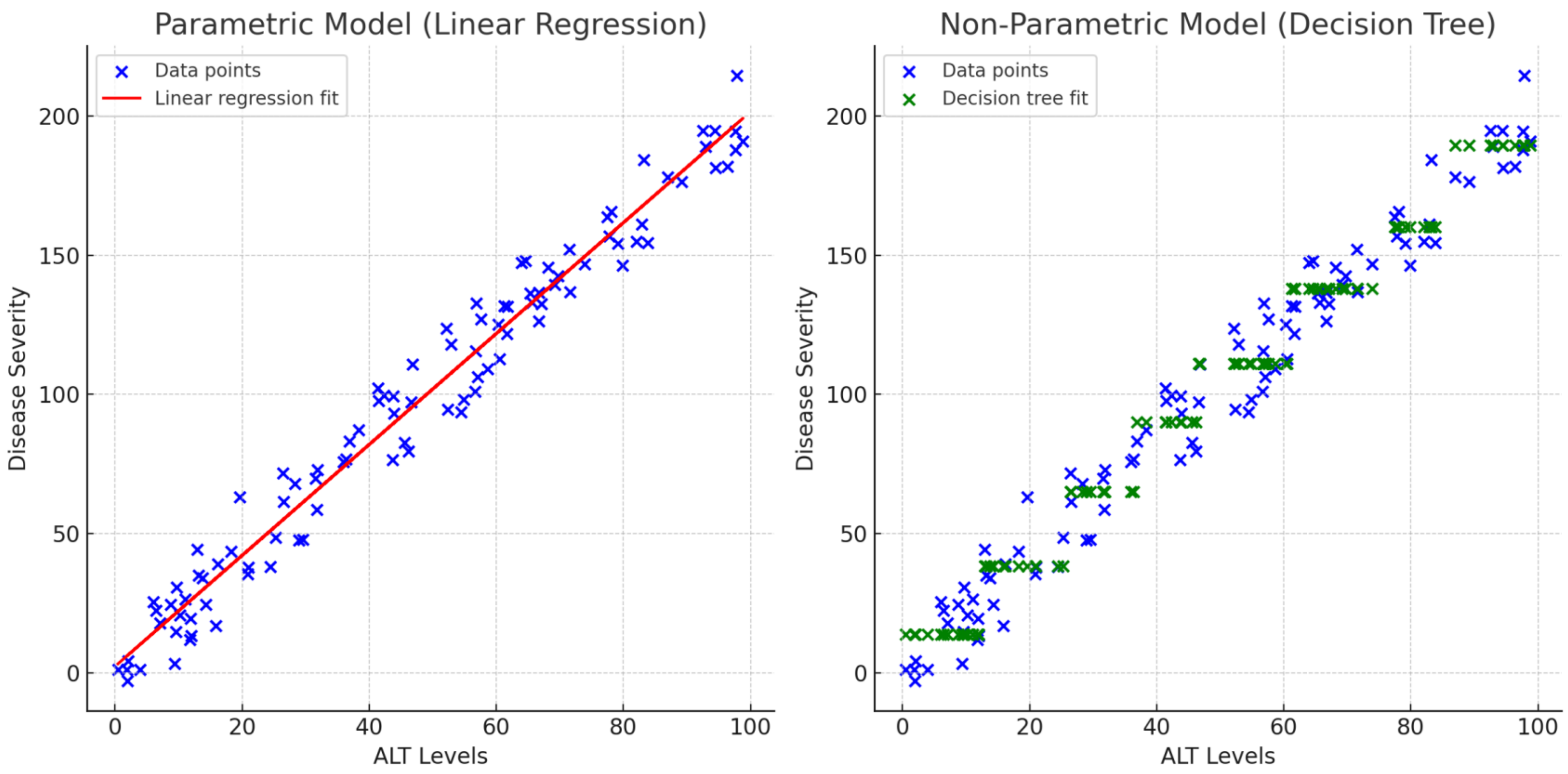
(x_n,y_n)

$$Y \approx f(X)$$

How to estimate $f(X)$?

- X
  - Independent variable(s), inputs, predictors, features
    - Continuous, categorical or mixed
- Y
  - Dependent variable, output, response, outcome
    - **Regression** if Y is continuous
    - **Classification** if Y is binary
- Parameters of $f(X)$

How to estimate *f*(*X*)?



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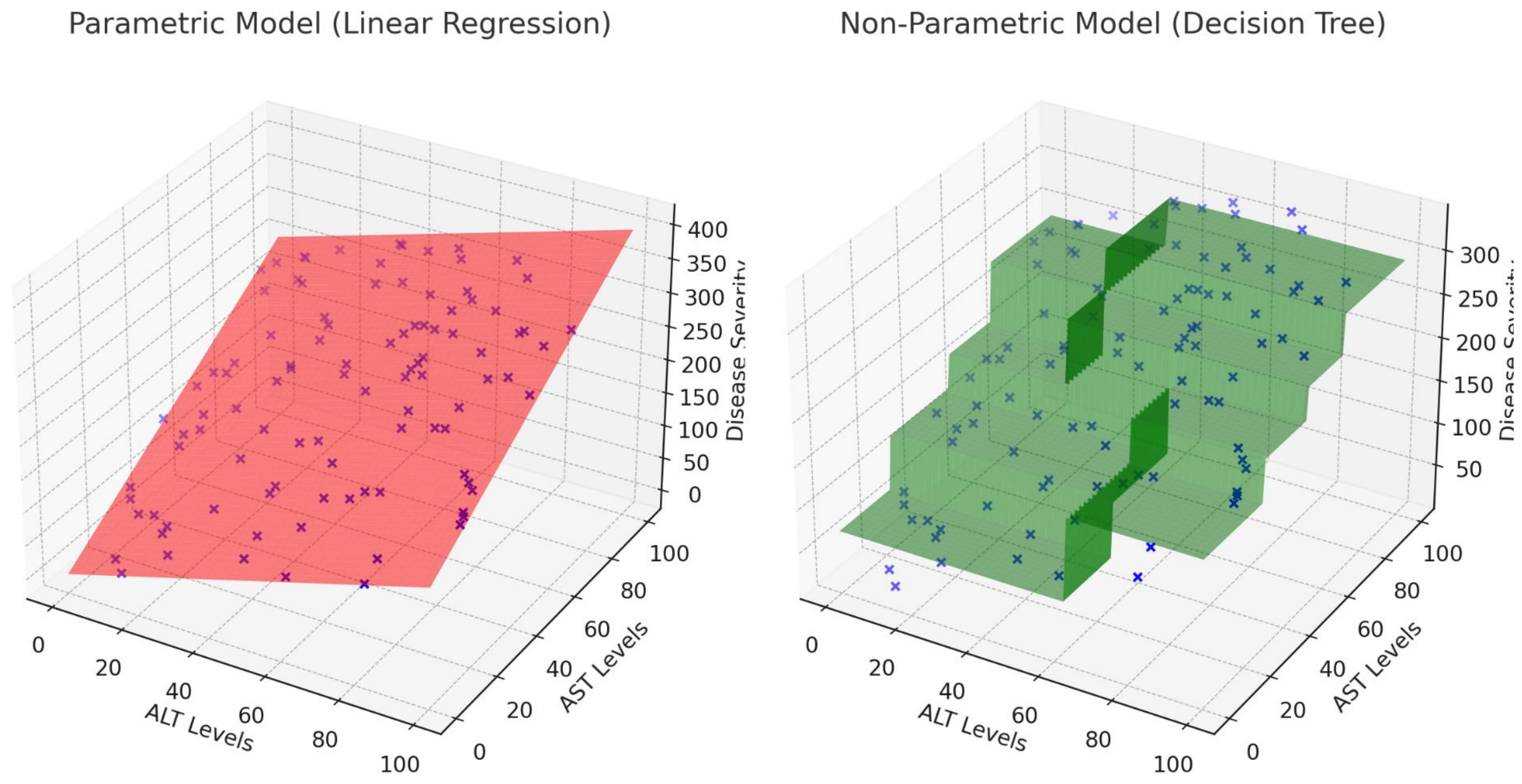
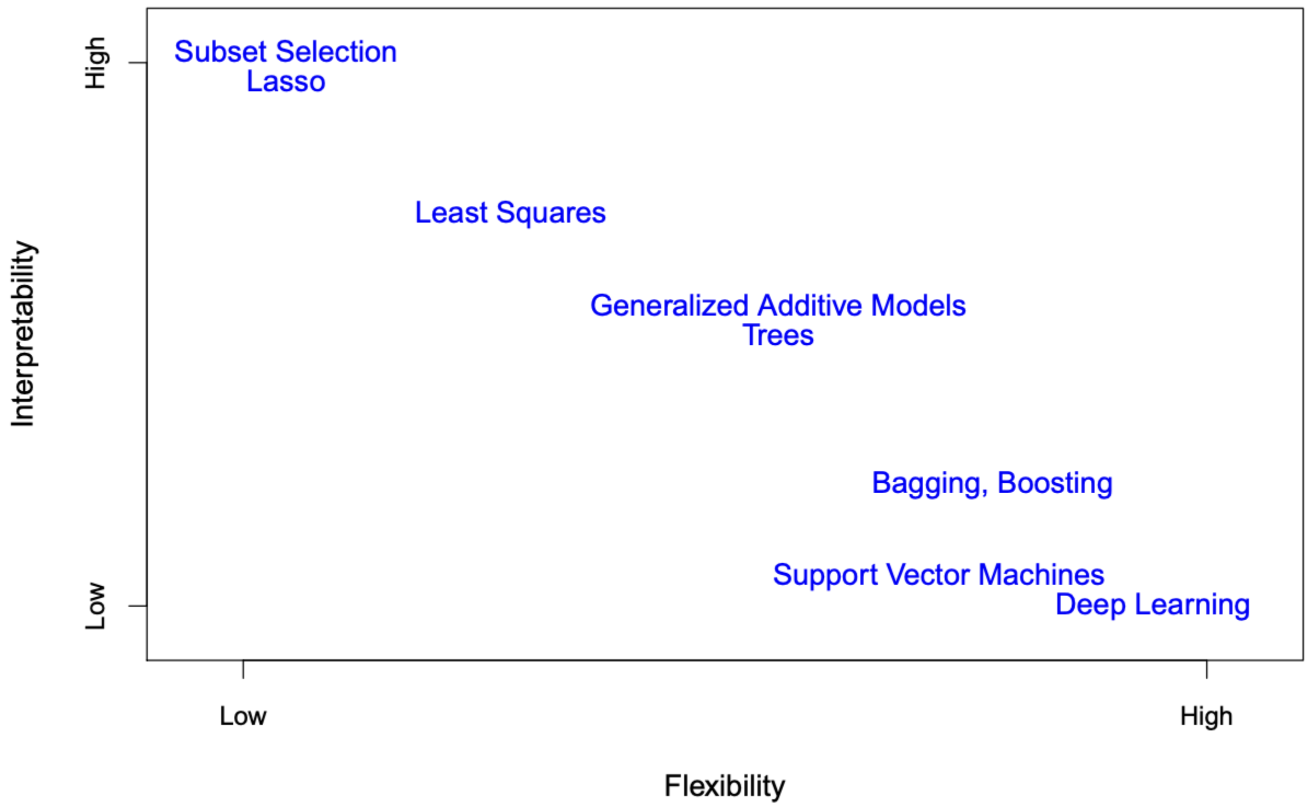
How to estimate *f*(*X*)?



How to estimate *f*(*X*)?



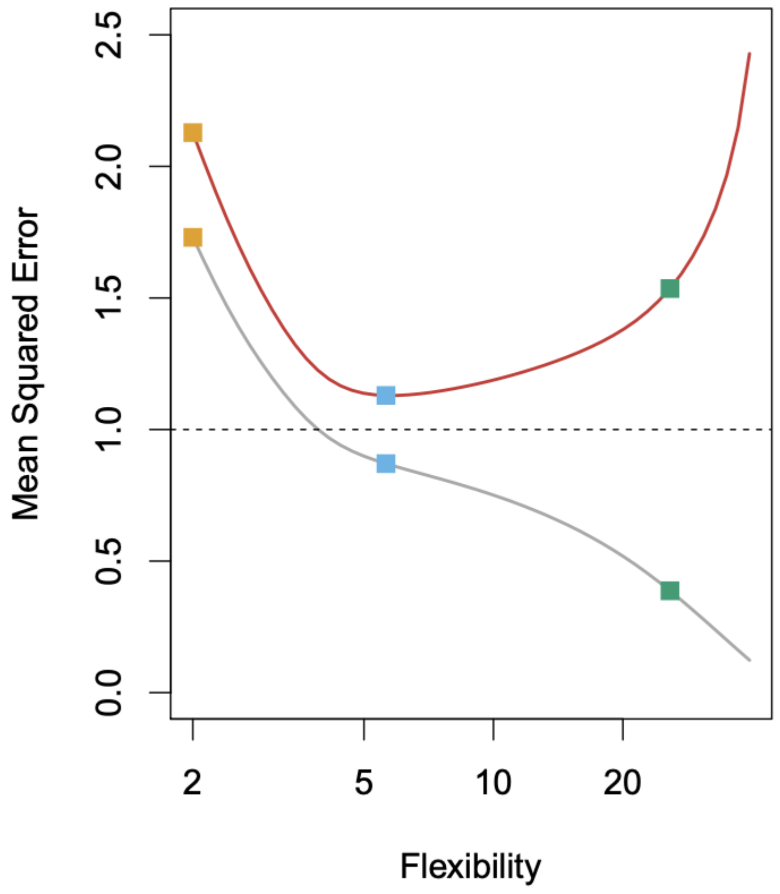
How to evaluate estimation of *f*(*X*)?

We assess the error

$$Y \approx f(X) \quad \Rightarrow \quad J(y, \hat{y}) = MSE = \frac{1}{N}\sum_{n=1}^{N}(y_n - \hat{y}_n)^2$$

Cost function

How to evaluate estimation of *f*(*X*)?

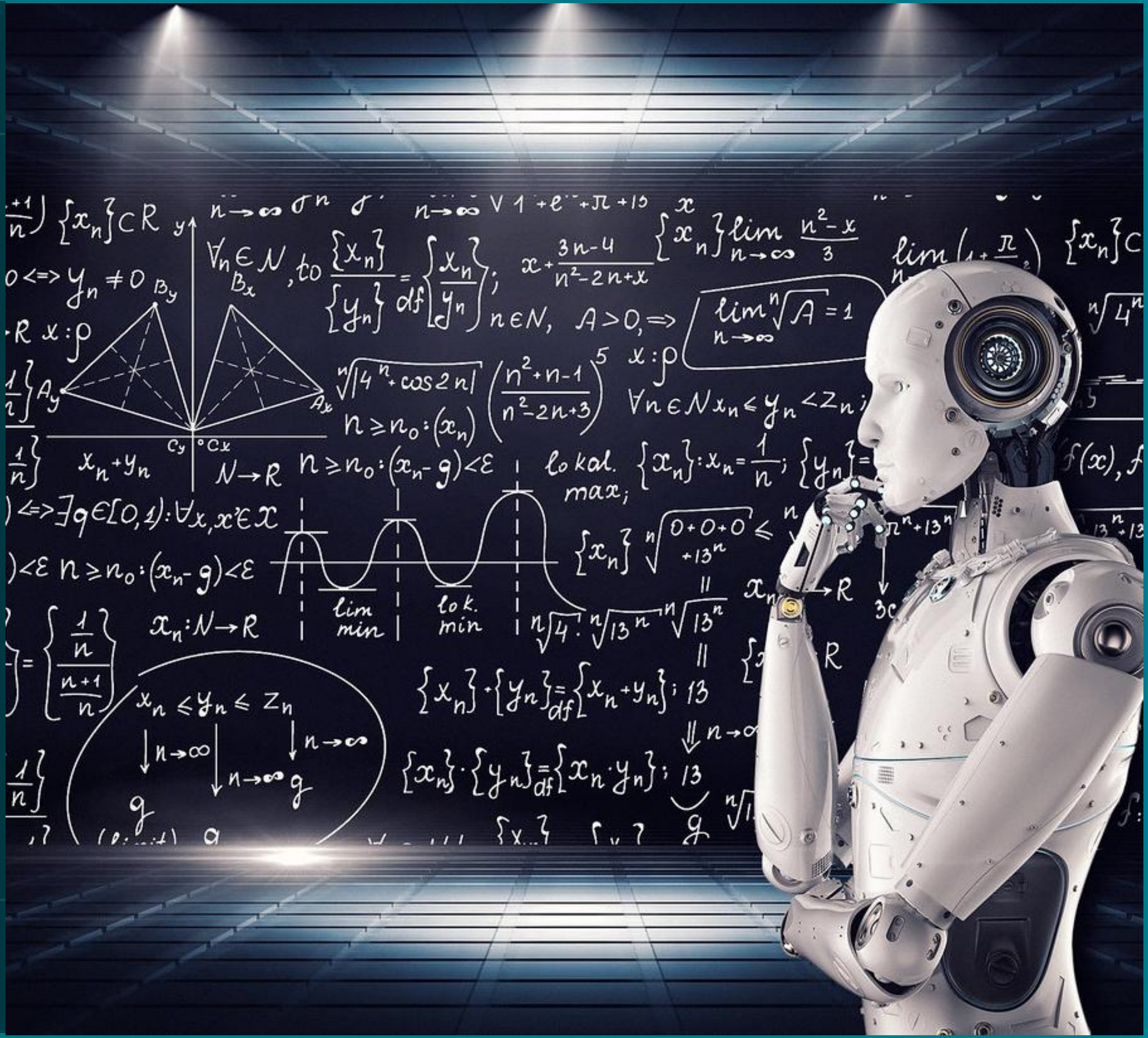


Testing set

Training set

Introduction to Machine Learning

Linear regression



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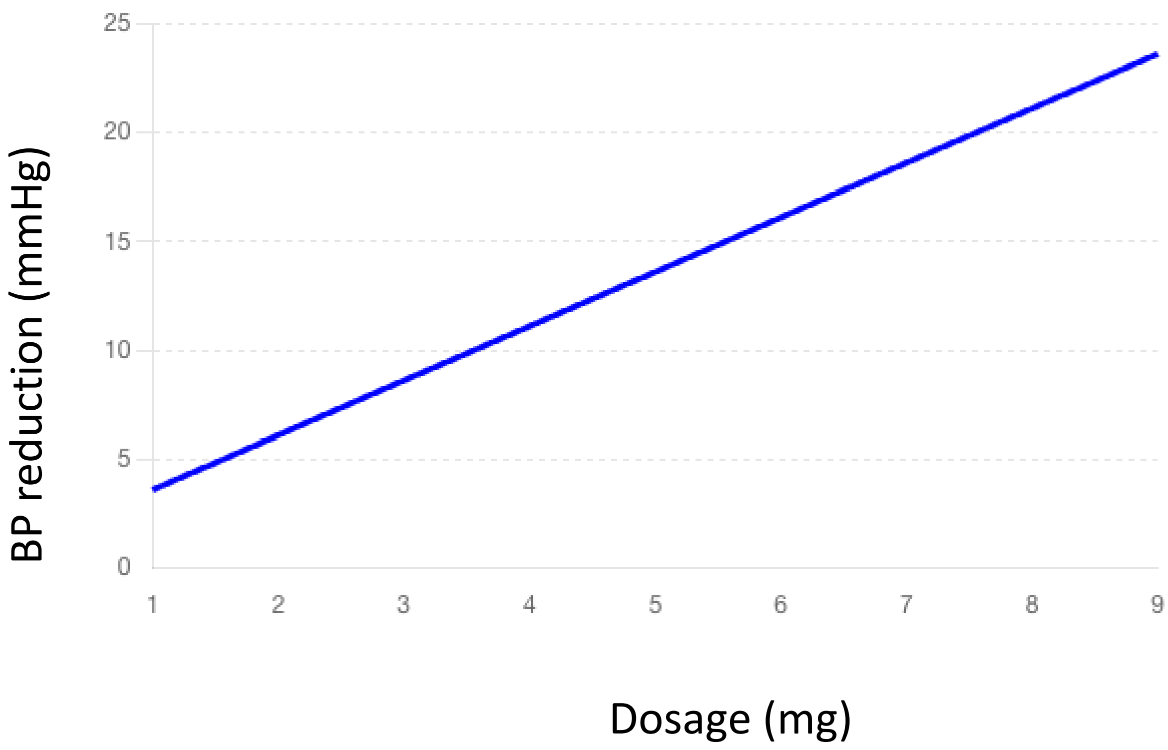
Simple Linear regression

$$\hat{Y} \approx \beta_0 + \beta_1 X + \varepsilon$$

Linear regression

$$\hat{Y} \approx \beta_0 + \beta_1 X$$

BloodPressureReduction = 4.1 + 2.5 * Dosage



Linear regression key assumptions

- **Linearity**: relationship between X and Y is linear
- **Independence**: observations are independent
- **Homoscedasticity:** constant variance of errors
- **Normality**: errors are normally distributed

Evaluating a Linear regression: assessing model's fit

- Mean Square Error (MSE)
  - MSE is the average of the squared differences between the actual and the predicted values

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Lower MSE indicates a better fit

- R-squared (R^2)
  - Measures the proportion of the variance in the dependent variable that is predictable from the independent variable(s)

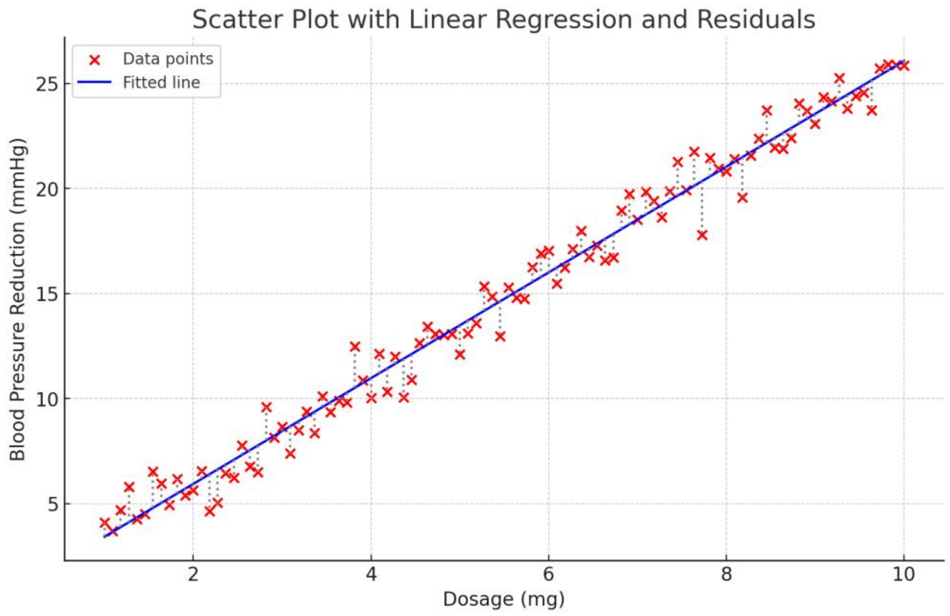
$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

Higher R^2 indicates a better fit

Fitting data: ordinary least sequares (OLS)

Concept: minimize the sum of squared residuals

- Analytical solution (exact answer)
- Fast computation
- Well-understood statistical properties



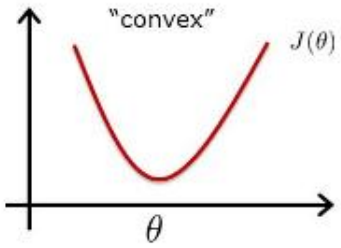
Goal: minimize $\mathrm{SSR} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} (y_i - (\beta_0 + \beta_1 X_i))^2$

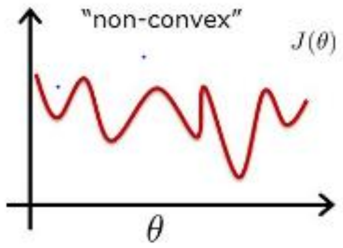
$\beta_1 = \Sigma(xi - \bar{x})(yi - \bar{y}) / \Sigma(xi - \bar{x})^2$
$\beta_0 = \bar{y} - \beta_1\bar{x}$

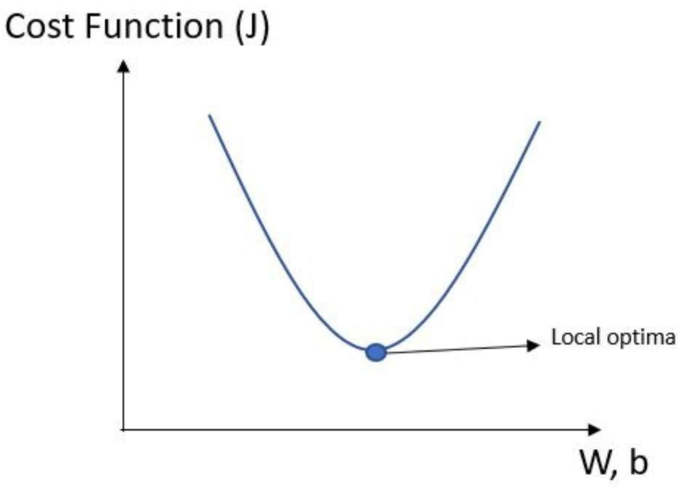
Fitting data: Gradient Descent

Concept: Iteratively adjust parameters to minimize cost function

- Large datasets where OLS is expensive
- When exact solution isn't available
- Foundation for more complex algorithms







$$\frac{dJ}{d\beta_0} = -\frac{1}{N}\sum_{i=1}^{N}[2(y_i - \hat{y}_i)]$$

$$\frac{dJ}{d\beta_k} = -\frac{1}{N}\sum_{i=1}^{N}[2(y_i - \hat{y}_i)x_{k_i}]$$

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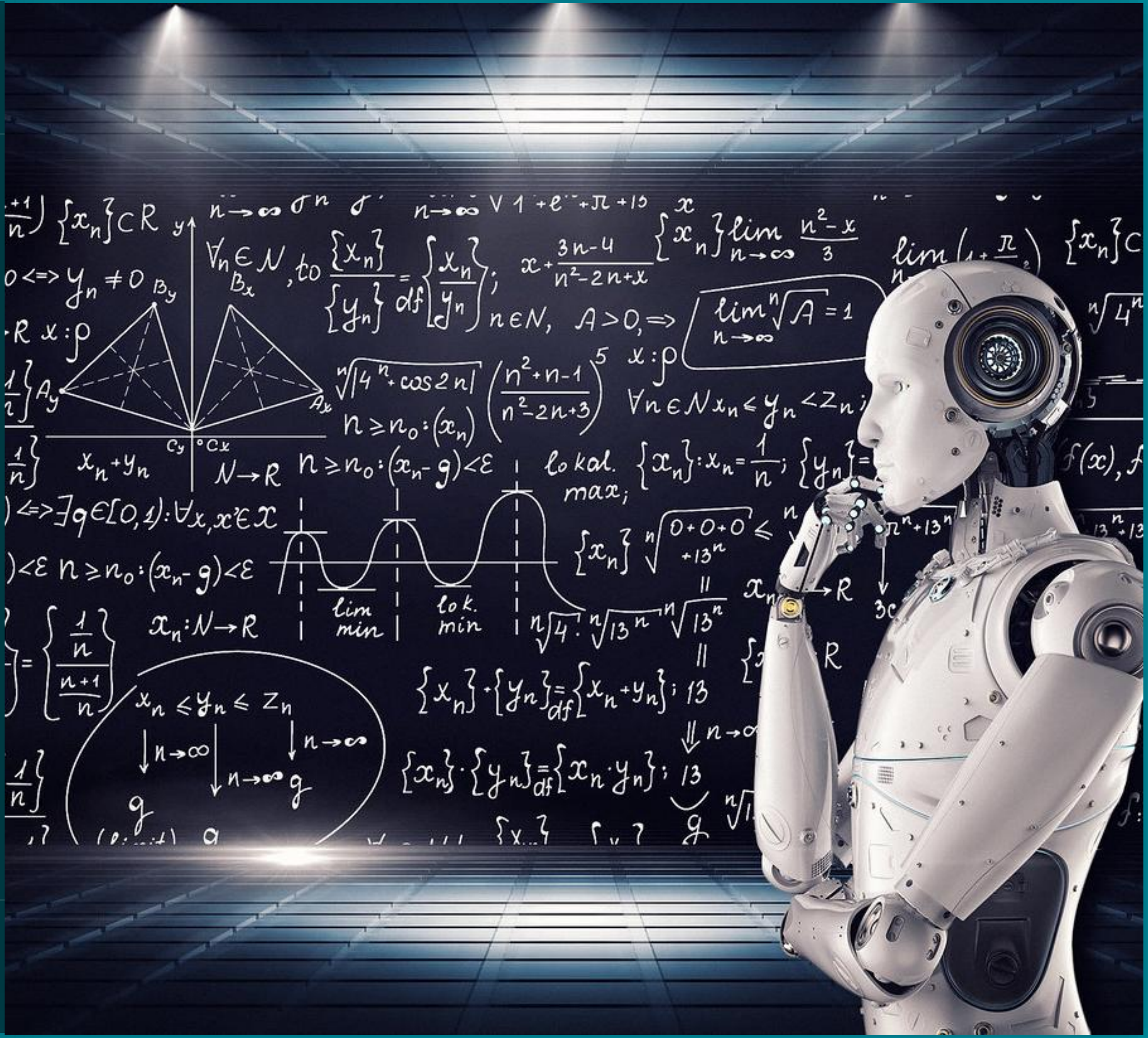
Multiple Linear regression



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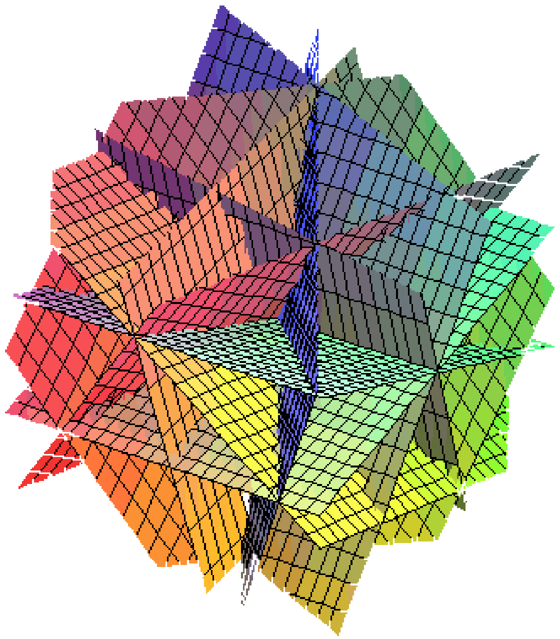
Multiple linear regression

$$\hat{y} = \beta_0 + \beta_1 \cdot x_1 + \beta_2 \cdot x_2 + \beta_3 \cdot x_3 + \beta_4 \cdot x_4 + \beta_5 \cdot x_5$$



$$\hat{y} = \beta_0 + \sum_{i=1}^{N} \beta_i \cdot x_i$$

Evaluating a multiple linear regression

- Mean Square Error (MSE)
  - MSE is the average of the squared differences between the actual and the predicted values

$$J(y, \hat{y}) = MSE = \frac{1}{N}\sum_{n=1}^{N}(y_n - \hat{y}_n)^2$$

Lower MSE indicates a better fit

- R-squared (R^2) or coefficient of determination
  - Measures the proportion of the variance in the dependent variable that is predictable from the independent variable(s)

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

Higher R^2 indicates a better fit

Linear regression: recap

- **Pros**
  - Really easy to understand (comparing to other algorithms)
  - Fast optimization (comparing to other algorithms)
  - Easy to extend the model (see next lessons)

- **Cons**
  - Sensible to outliers
  - Assumes that there is no multicollinearity
  - Feature scaling is required
  - Monotonicity assumption: for the model, the relation between each feature and the output

Other approaches to regression

Method	When to use	Pros	Cons
Linear	Linear relationships	Simple, interpretable	Assumes linearity
Polynomial	Curved relationships	Still interpretable	Can overfit easily
Regularization (Ridge/Lasso)	Many features, multicolinearity	Prevents overfitting	Less interpretable
Trees	Complex non-linear patterns	No assumptions needed	Can overfit, less stable
Neural networks	Vary complex patterns, lots of data	Very flexible, automatic feature learning	Needs lots of data, black box, computation